BIOS 6618 Midterm Project

Due by Friday, October 24, 2025 by 11:59 pm to Canvas Assignment Basket

Overview

This project is intended to give you practice with conducting a "rigorous" simulation study to examine some question or problem of interest. We have been applying simulations in lecture, lab, recitation, and homeworks throughout the semester. However, these examples are very targeted and brief. In this project you will work on a simulation problem that you propose, and then write a brief two-page summary report.

For your chosen topic, you'll prepare a 2-page summary (strict maximum; single spaced; at least 1 figure and 1 table must be in the 2-page limit; references can extend beyond the 2-page limit) in the form of a brief scientific report. Page 2 includes an example outline with potential lengths for different sections to consider for your simulation study report.

The rubric that will be used for grading is posted on our Canvas Midterm Project assignment page. In general, you will be expected to write in the style of a journal article/brief scientific report (i.e., avoiding overly casual language and overuse of "I/we feel"). Refer to the "Journal of BIOS 6618 Author Guidelines" HTML document for additional formatting requirements.

Some Potential Ideas:

- Sample size calculation for a study where we do not have an easily accessible power calculator (e.g., non-inferiority study designs, equivalence study designs, risk ratio, chi-squared test, permutation t-test)
- Theorem evaluation (e.g., central limit theorem under various distributions and sample sizes, other theorems you've seen in papers/textbooks/classes)
- Compare methods (e.g., the exact versus asymptotic approaches for a given method, performance of a t-test vs. permutation t-test vs. chi-squared test either for binary data or by coercing a continuous variable, compare asymptotic confidence intervals from a function's output with bootstrap percentile confidence intervals)
- Approximations for distributions (e.g., the Poisson approximation to the binomial)
- Taking any model or approach we've discussed and evaluating what happens when we violate our assumptions (e.g., how poorly does the t-test perform across various measures if we do not have normally distributed data)

A few thoughts on simulation studies:

- Remember, you get to set the "truth" that you are simulating from! You have the power to make things very subtle or extremely drastic.
- When choosing scenarios, it can be helpful to fit one that meets the assumptions that you can compare to (e.g., simulating normal data for a t-test to compare to exponentially simulated data with the same mean). If you are comparing to a known theory (i.e., you have a reference or what should happen theoretically), you may be able to just reference that as your comparator. Although, it can still be useful (and fun) to confirm the theoretical properties via simulation!
- Try to only "break" one part of the problem so you can better attribute why you are observing your results. If you change 3 things at once, it can be hard to know what specific choice on its own led to what you observe.
- The "outcomes" you measure will depend on the question you are exploring via simulation. They may include the mean, median, bias, standard deviation, mean square error, relative efficiency, coverage, power, type I error rate, etc.

Example Outline of Brief Scientific Report/Summary for Simulation Studies

Introduction with brief background – 0.25 page

Question to explore, motivating problem, theorem description – 0.25 page

Simulation set-up – 0.25 page

Results – 1 page
- At least one plot
- Tables summarizing either the simulation scenario(s) and/or results

Discussion and Conclusion – 0.25 page

References (not included in 2-page limit)
- You may not have any references for this project and that is okay.

Appendix for Reproducible Research Goal (not included in 2-page limit):
- R code with clear comments

Note, these examples and resources were originally identified for the Final Project that involves an applied data analysis, but may still be helpful in this shorter simulation study project:

Refer to example: AJPH Example Brief Report.pdf, located on the Final Project Canvas page

General Writing Resource: Tips on Writing Results for a Scientific Paper Amstat News - Cummins - Sept 2009.pdf, located on the Final Project Canvas page

Reproducible Research

You will need to follow Reproducible Research principles in carrying out your project. Adherence to the principles of reproducible research is essential to the reporting of scientific results.

Be sure to familiarize yourself with some of the resources on this topic listed below.

An outstanding example of reproducible research by Keith Baggerly and Kevin Coombes of the MD Anderson Cancer Center Bioinformatics and Computational Biology department is posted in the Paper Repository ("Deriving Chemosensitivity from Cell Lines: Forensic Bioinformatics and Reproducible Research in High-Throughput Biology" in the *Annals of Applied Statistics* by Baggerly and Coombes, and *Cancer Letter* 1, 2, and 3 by Paul Goldberg).

RR Resources: (Canvas Paper Repository)

Gentleman and Temple Lang paper – Statistical Analysis and Reproducible Research; from the Bioconductor Project

Roger Peng 2009 Editorial in *Biostatistics*

Roger Peng 2011 article in *Science*

Frank Harrell presentation on the use of R in clinical trials research

IOM Report on Evolution of Translational Omics

Lehrer 2010 New Yorker The Truth Wears Off

Dynamic/computable documents using R
http://yihui.name/knitr/
http://rmarkdown.rstudio.com/

Dynamic/computable documents using SAS
http://sites.northwestern.edu/stattag/

Git and GitHub – version control, collaboration tools
https://git-scm.com/
https://github.com/

Recommended, but not required:

Use one of these (or another) computable document applications.